



SEMESTER 2 EXAMINATIONS 2024/2025

MODULE: CSC1111 - Building Complex Computational Models

PROGRAMME(S):
DS BSc Data Science

STAGE: 4

EXAMINER(S):

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TIME ALLOWED: 3 Hours

INSTRUCTIONS: Answer 4 questions. All questions carry equal marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.

The use of programmable or text storing calculators is expressly forbidden.

Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

QUESTION 1

[TOTAL MARKS: 25]

Q 1(a)

[13 Marks]

- (i) Explain carefully what is meant by rendering a time series (a) trend stationary and (b) difference stationary. [4 Marks]
- (ii) For the graph in Figure Q1(a), explain how one of these techniques may be used to render it stationary. Draw a rough sketch of the result of applying such a technique to the result. [5 Marks]
- (iii) Draw rough sketches of the Auto Correlation Function (ACF) of the function in Figure Q1(a) and of the detrended graph of it. [4 Marks]

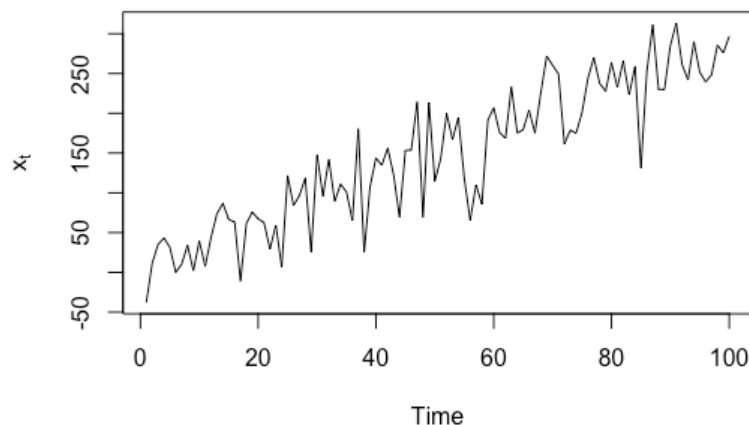


Figure Q1(a)

Q 1(b)

[4 Marks]

Explain clearly and concisely what you understand by the terms *Auto-Correlation Function (ACF)* and *Partial Auto-Correlation Function (PACF)*.

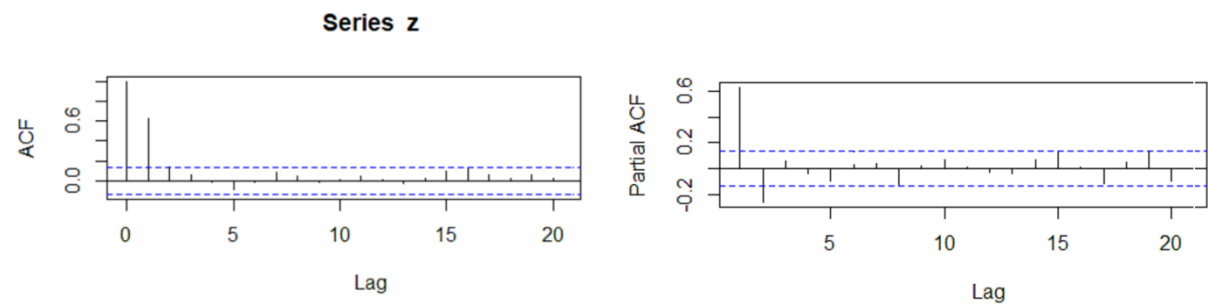
Q 1(c)

[8 Marks]

Figure Q1(c)(i) and (ii) show 2 examples of (theoretical) Autocorrelation Function diagrams for some ARIMA and SARIMA models, respectively. In each case, give the simplest possible form you can of the underlying model. You should justify your choice in all cases.

Case (i) Series z: ACF and PACF

[3 Marks]



Case (ii) Series z: ACF and PACF

[5 Marks]

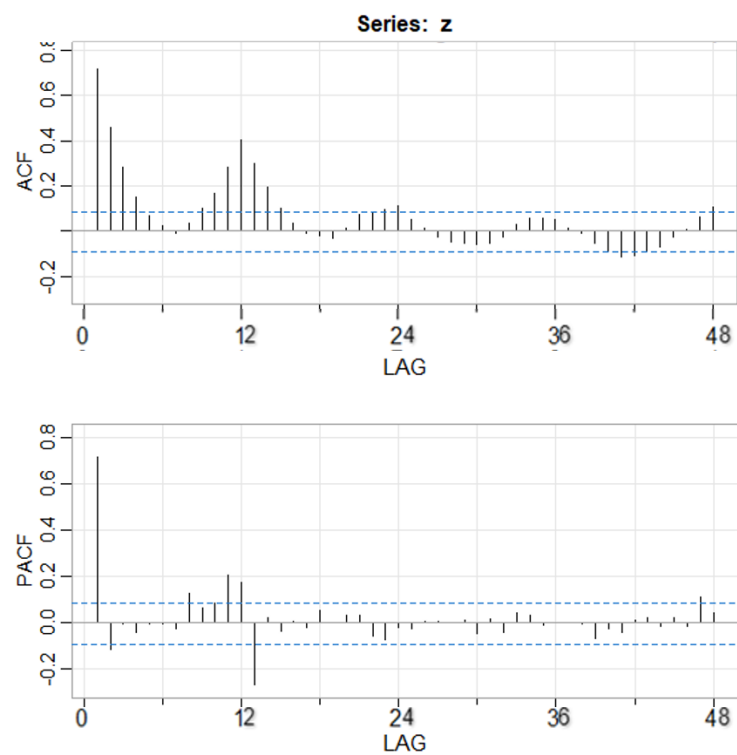


Figure Q1(c)

[End of Question 1]

QUESTION 2

[TOTAL MARKS: 25]

Q 2(a)

[4 Marks]

As part of the course, define carefully in words and by means of a rough figure what are the four time series components that we have covered?

Q 2(b)

[6 Marks]

The time series graph in Figure Q2(b) shows a plot of Monthly Retail Sales from New South Wales Department Stores from 1982-2001 with part of the plot enlarged to show detail. Identify the two time series components that you can observe in this plot. Identify, giving reasons, the possible causes of these components, given that the data represents Monthly Retail Sales.

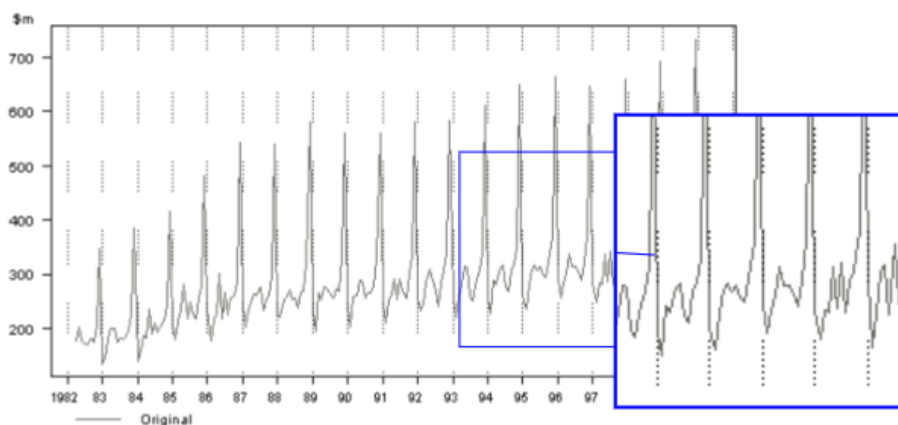


Figure Q2(b)

Q 2(c)

[15 Marks]

On the course, we learned about two types of models that a time series can be decomposed into. The time series data in Table Q2(c) shows Quarterly Sales Data (in thousands) from 1979-81. Generate a graph of the data explaining what kind of model should be used to decompose the data and provide reasons for your explanation. Calculate the Trend (using 2X4pt Moving Averages), Seasonal Index and Seasonally-Adjusted data. Graph the Trend and Seasonally-Adjusted data (but **not** the Seasonal Index data).

Year	Quarter	Sales
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1979	1	72
1979	2	110
1979	3	117
1979	4	172
1980	1	76
1980	2	112
1980	3	130
1980	4	194
1981	1	78
1981	2	119
1981	3	128
1981	4	201
1982	1	81
1982	2	134
1982	3	141
1982	4	216

Table Q2(c)

[End of Question 2]

QUESTION 3**[TOTAL MARKS: 25]****Q 3(a)****[7 Marks]**

In terms of forecasting, define carefully the difference between an *ex post* and an *ex ante* forecast. Suppose you have training data and you keep a subset of that data to test a pretrained model. Which type of forecasting would you say this is? Give a reason for your answer.

Q 3(b)**[12 Marks]**

The population of the American (Plains) bison (in 1,000s) is shown in Table Q3(b). As can be seen, the population of the bison has been dropping for a number of decades. You are required to forecast the population in 2020 using the values shown using Exponentially Weighted Moving Averages (EWMA) with smoothing values of 0.2, 0.5 and 0.8. Please sketch a graph of the population and forecast values for different smoothing values.

	Population x 1000
Year	
1920	1770
1930	1704
1940	1658
1950	1852
1960	1698
1970	1954
1980	1782
1990	1660
2000	1602
2010	1650

Table Q3(b)

Q 3(c)

[6 Marks]

Given that the values of the population of the American (Plains) bison, shown in Table Q3(b), have been dropping for some decades, ecologists are required to take some action. In your opinion, which of the smoothing average values should they use to reach a decision on this action? Give a reason for your answer.

[End of Question 3]

QUESTION 4

[TOTAL MARKS: 25]

Q 4(a)

[8 Marks]

Give your own explanations and examples of:

- i) Emergent behaviour
- ii) Self-organisation

What is the difference between the two?

Q 4(b)

[9 Marks]

What is considered a good model in complex systems? Give your own example of a good model design explained through typical modelling steps.

Q 4(c)

[8 Marks]

The figure Q4(c) shows the bifurcation diagram of a parameter θ . Please give clear explanations for what is happening at critical points for θ (e.g. for θ values around ~ 3.5 and ~ 14).

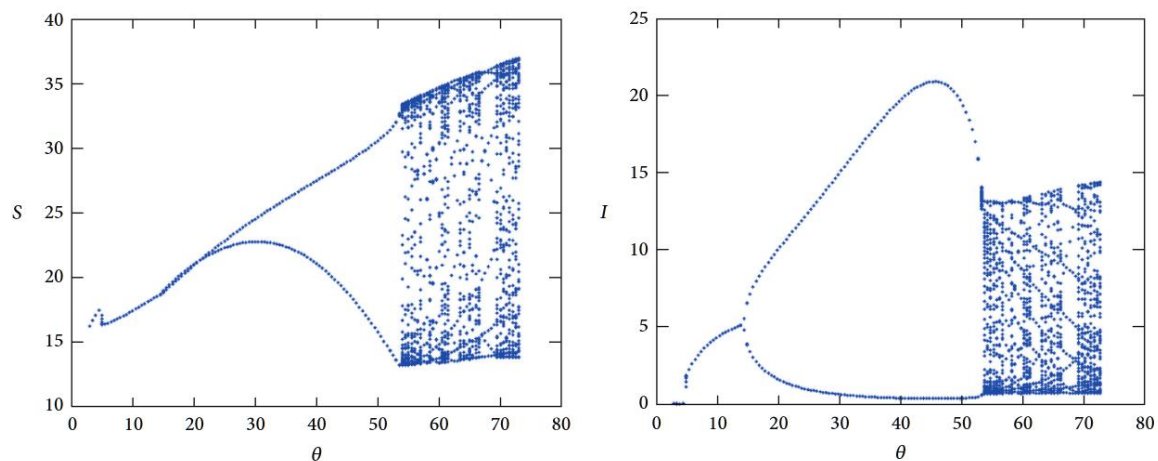


Figure Q4(c)

[End of Question 4]

QUESTION 5**[TOTAL MARKS: 25]****Q 5(a)****[6 Marks]**

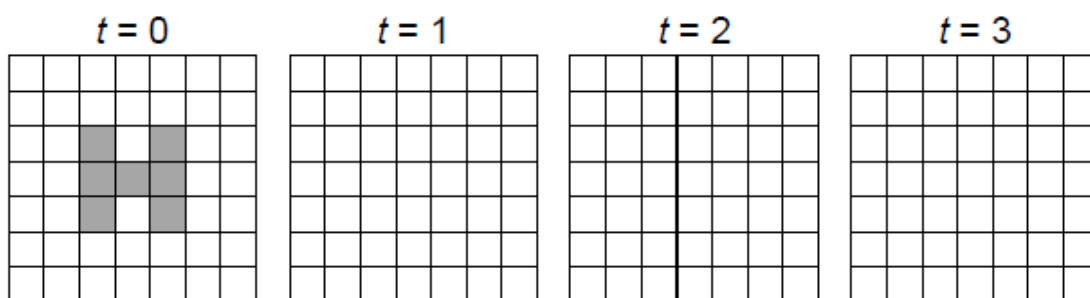
- (i) What are the typical properties of agents in Agent-Based Modelling (ABM)?
- (ii) Using an example of birds flocking, draw main rule sets showing how they can be used in ABM. (You can use similar logic as e.g. in boids example).

Q 5(b)**[5 Marks]**

Using a method of your choice (e.g. Louvain method, Girvan-Newman method, random walk method or other) explain concisely how it can be used to find optimal community structure that maximises the modularity. Explain what are the advantages of using one method over the other?

Q 5(c)**[6 Marks]**

Figure Q5(c) is an example of a 2D Cellular Automata (CA) model with von Neumann neighbourhood and with *no boundary* (infinite space). White means 0, gray means 1. Each cell switches to $\text{round}(S/5)$ at every time step, where S is the local sum of states within its neighbourhood. Complete the time evolution of this CA.

**Figure Q5(c)****Q 5(d)****[8 Marks]**

Using a simple 2D forest fire example model (or another such application from your knowledge), draw a schematic illustration of the model's state-transition function. Clearly define all possible states and write what typical properties of CA can be seen in this example.

What would happen if the space of the forest fire propagation were 1-D or 3-D?

[End of Question 5]

QUESTION 6**[TOTAL MARKS: 25]****Q 6(a)****[4 Marks]**

What is a Markov Process? What are the main assumptions of a Markov Process?

Q 6(b)**[8 Marks]**

A video game character's health status can be classified into four levels: Healthy (H), Injured (I), Critical (C), and Near-Death (N). After each combat encounter, the character's health status may change according to certain probabilities.

- Assume the following transition probabilities from the Healthy state: Injured (0.75), Critical (0.2), Near-Death (0);
- Assume the following transition probabilities from the Injured state: Healthy (0.25), Injured (0.3), Critical (0.45);
- Assume the following transition probabilities from the Critical state: Injured (0.4), Critical (0.3), Near-Death (0.2);
- Assume the following transition probabilities from the Near-Death state: Injured (0.15), Critical (0.3), Near-Death (0.55).

- (i) Draw a Markov transition diagram and associated transition matrix for this system.
- (ii) From your diagram in (i), find the probability that an Injured character will transition to Healthy after 1, 2 combat encounters.

Q 6(c)**[5 Marks]**

Explain concisely what is meant by a Markov Chain Monte Carlo (MCMC) and when it should be used.

Q 6(d)**[8 Marks]**

Explain MCMC general algorithm and workflow in terms of prior and posterior belief. Specifically demonstrate what is the role of a Sampler in MCMC algorithm (choose a Sampler of your choice highlighting its advantages).

[End of Question 6]**[END OF EXAM]**